AI Study Assistant

A lightweight document summarization and question-answering assistant

Workplace AI | Battlefield Thunderstorm

Stage I proposal • October 2026

01 / Problem & motivation

Students spend time navigating lengthy English course readings.

Key information is scattered across PDF handouts and notes.

A simple document assistant can reduce the effort needed to locate relevant passages.

02 / Proposed solution

Upload an English PDF or TXT document.

Extract representative sentences as a short summary.

Ask a question and retrieve relevant source passages.

Future extension: locally hosted generative AI with citations.

03 / Stage I working prototype

Python + Gradio web interface

PDF text extraction with pypdf

Sentence-frequency extraction for summaries

TF-IDF similarity for passage retrieval

Offline, no API key, no model training required

04 / Software architecture

User uploads PDF / TXT

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Text extraction and sentence segmentation

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Summarization scoring / TF-IDF retrieval

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Gradio displays summary or relevant source excerpts

05 / Hardware roadmap

Current Stage I demo: runs on a standard laptop CPU.

Target competition platform: Intel Arc Pro B70 (32 GB).

Stage II proposal: integrate a local language model and investigate accelerated inference.

Hardware compatibility, latency, and accuracy have NOT yet been validated.

06 / User experience

Step 1 Upload a text-based document.

Step 2 Click “Extract summary” to see key sentences.

Step 3 Enter a question in “Ask document.”

Step 4 Read relevant excerpts directly from the source.

07 / Expected value & use cases

Students: quickly review English study materials.

Educators: explore course handouts with simple retrieval.

Privacy: local baseline does not send documents to a remote API.

Accessibility: lightweight Python prototype with minimal setup.

08 / Evaluation & limitations

Potential metrics: passage relevance, summary coverage, task completion time.

Current baseline is extractive, not a generative LLM.

No OCR for scanned PDFs; English-only heuristics.

No quantitative user study or hardware benchmarks have been conducted.

09 / Future work

Add an on-device language model with evidence-grounded answers.

Evaluate answer faithfulness and document retrieval quality.

Optimize inference for Intel Arc Pro B70 after hardware access.

Test representative student documents and measure latency.

10 / Team & submission links

Participant: [YOUR FULL NAME]

Student ID: [YOUR STUDENT ID]

GitHub Repository: [PASTE YOUR PUBLIC REPOSITORY URL]

Demo Video (YouTube, ≤ 3 minutes): [PASTE VIDEO URL]

Replace all placeholders before submission.